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Cancer Rehabilitation

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SUMMARY PEARLS

- Radiation therapy is a common treatment for cancer; however, both the tumor and the surrounding healthy tissue can sustain damage from the radiation, leading to both direct and indirect injury, including late-effect radiation fibrosis syndrome.
- Chemotherapy agents can cause significant normal tissue toxicity and complications, such as peripheral neuropathy, cognitive dysfunction, and cardiotoxicity.
- Aromatase inhibitors used in adjuvant endocrine therapy can cause side effects, including osteoporosis, increased fracture risk, weight gain, and various musculoskeletal symptoms.
- Cancer-related fatigue is a common symptom affecting patients at all stages of their disease and can persist for months to years, significantly impacting function and quality of life. Despite its high prevalence, it is often underrecognized and undertreated.
- Cancer-related cognitive impairment is common during active cancer treatment and characterized by difficulties with executive function, concentration, memory, processing speed, and short-term memory.
- Lymphedema results in localized swelling that can cause physical symptoms, functional limitations, and negative impact on quality of life.
- Chemotherapy-induced peripheral neuropathy is a common adverse effect resulting from injury to peripheral nerves during cancer treatment.
- The treatment of bone metastases can include medication, radiation therapy, and/or surgical interventions, and rehabilitation can be beneficial but requires precautions.

Introduction

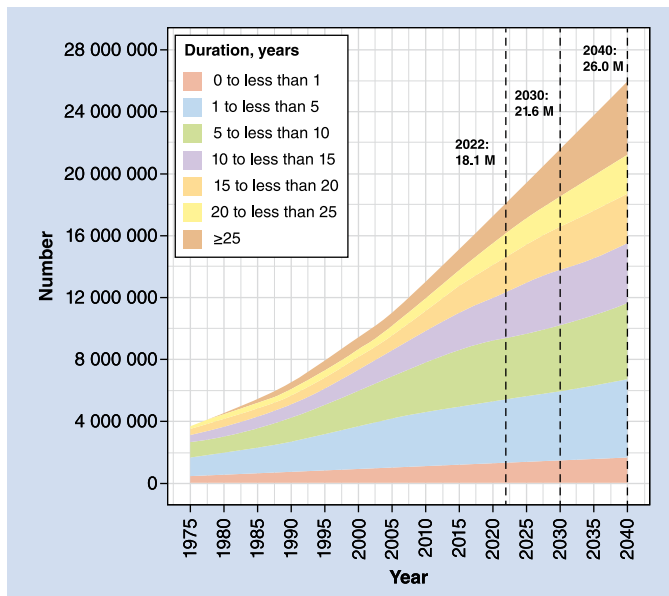
The prevalence of cancer survivors has increased over the past decades and is anticipated to continue to increase (Fig. 30.1).¹⁶⁵ The National Cancer Institute categorizes any living individual who has been diagnosed with cancer, excluding skin cancers, as a cancer survivor. The majority of this large and growing population of cancer survivors will experience symptoms or functional impairments related to their cancer and/or its treatment, which can deleteriously impact their function and quality of life.¹¹⁸ The ramifications of cancer and its treatment can be long term and progressive. This expanding population highlights the need for the increasing awareness and expertise of physiatrists to provide specialized rehabilitation care for cancer survivors to optimize their function and quality of life.

Cancer rehabilitation aims to help cancer survivors regain their highest possible level of functionality and quality of life.⁸³ A succinct definition of the scope of cancer rehabilitation is: "Cancer rehabilitation is medical care that should be integrated throughout the oncology care continuum and delivered by trained rehabilitation professionals who have it within their scope of practice to diagnose and treat patients' physical, psychological, and cognitive impairments in an effort to maintain or restore function, reduce symptom burden, maximize independence, and improve quality of life in this medically complex population."¹⁴¹ In addition to direct effect from cancer and its treatment, additional impairments

resulting from coexistent diseases, such as ischemia and arthritis, commonly coexist. Their effective rehabilitation requires thoughtful attention to cancer-specific concerns. These may include life expectancy, individualized goals, symptom burden, and other treatment-related side effects. All these factors need to be considered when crafting compassionate and individualized treatment plans. This chapter provides an understanding of key issues relevant to the rehabilitation of cancer survivors, with a particular focus on impairments related to a nervous system and/or musculoskeletal etiology. This will include a brief introduction and overview of cancer treatment modalities, followed by reviewing common symptoms experienced by cancer survivors, highlighting specific populations of cancer survivors, and concluding with a review of precautions to ensure safety with the rehabilitation approach.

Epidemiology

Cancer is a common condition, impacting millions of Americans, and the number of cancer survivors is projected to reach 26 million by 2040.¹⁶⁵ Breast cancer is the leading cancer diagnosis in females and prostate cancer in males. These are followed by lung cancer and then colorectal cancer for both sexes.⁴ The leading sites of new cancer diagnosis and death by sex are highlighted in Table 30.1.



• **Fig. 30.1** Growing population of cancer survivors living in the United States by year and time (in years) since diagnosis. (From Tonorez E, Devasia T, Mariotto AB, et al. Prevalence of cancer survivors in the United States. *J Natl Cancer Inst.* 2024;116(11):1784-1790.)

TABLE 30.1A 2024 Estimates for Leading Sites of New Cancer Cases in Males

Prostate	299,010	29%
Lung	116,310	11%
Colorectal	81,540	8%
Bladder	63,070	6%
Melanoma	59,170	6%
Renal	52,380	5%
Non-Hodgkin lymphoma	44,590	4%
Oral cavity and pharynx	41,510	4%
Leukemia	36,450	4%
Pancreas	34,530	3%

Data from American Cancer Society. *Cancer Facts & Figures 2024*. American Cancer Society; 2024.

Disparities

Disparities are present within cancer incidence, treatment, and survival. Patients with lower socioeconomic status (SES) have a higher incidence as well as higher likelihood of dying.¹⁶⁵ This was particularly marked for lung, colorectal, cervical, stomach, and liver cancer.¹⁴² Even after adjusting for poverty rate, non-Hispanic White individuals have a higher 5-year survival rate compared to Black, American Indian/Alaskan Native, and Asian/Pacific Islander males, as well as Black and American Indian/Alaskan Native females.¹⁷⁵ The more recent rates by race and ethnicity reported by the American Cancer Society are noted in [Table 30.2](#). There are potential ways to decrease cancer disparities by focusing on prevention methods, such as reducing tobacco use, physical

TABLE 30.1B 2024 Estimates of Leading Sites for Cancer Deaths in Males

Lung	65,790	20%
Prostate	35,250	11%
Colorectal	28,700	9%
Pancreas	27,270	8%
Liver	19,120	6%
Leukemia	13,640	4%
Esophagus	12,880	4%
Bladder	12,290	4%
Non-Hodgkin lymphoma	11,780	4%
Brain and central nervous system	10,690	2%

Data from American Cancer Society. *Cancer Facts & Figures 2024*. American Cancer Society; 2024.

TABLE 30.1C 2024 Estimates for Leading Sites of New Cancer Cases in Females

Beast	310,720	32%
Lung	118,270	12%
Colorectal	71,270	7%
Uterine	67,880	7%
Melanoma	41,470	4%
Non-Hodgkin lymphoma	36,030	4%
Pancreas	31,910	3%
Thyroid	31,520	3%
Renal	29,230	3%
Leukemia	26,320	3%

Data from American Cancer Society. *Cancer Facts & Figures 2024*. American Cancer Society; 2024.

TABLE 30.1D 2024 Estimates of Leading Sites for Cancer Deaths in Females

Lung	59,280	21%
Breast	42,250	15%
Pancreas	24,480	8%
Colorectal	24,310	8%
Uterine	13,250	5%
Ovary	12,740	4%
Liver	10,720	4%
Leukemia	10,030	3%
Non-Hodgkin lymphoma	8360	3%
Brain and central nervous system	8070	2%

Data from American Cancer Society. *Cancer Facts & Figures 2024*. American Cancer Society; 2024.

TABLE 30.2 Morality Rates by Race and Ethnicity per 100,000 People From 2016-2020⁴

	All	White	Black	American Indians and Alaskan Natives	Asian American /Pacific Islander	Hispanic /Latino
All sites	149.8	155	175.8	183.8	95.4	108.6
Male	178	183.3	217.4	221.6	111.6	130.2
Female	129.1	133.6	150.2	157.9	83.7	93.5
Breast (female)	19.7	19.7	27.8	21.1	11.8	13.7
Colon and rectum	13.2	13.1	17.7	19	9.2	10.7
Male	15.7	15.5	22.4	23.1	11	13.6
Female	11	11.1	14.4	16	7.8	8.5
Kidney and renal pelvis	3.5	3.6	3.4	6.7	1.6	3.3
Male	5.1	5.3	5.2	9.9	2.4	4.8
Female	2.2	2.3	2.2	4.2	1.0	2.1
Liver and intrahepatic bile duct	6.6	5.9	8.3	13.6	8.5	9.3
Male	9.6	8.5	13	19.9	12.6	13.1
Female	4.1	3.7	4.8	8.8	5.2	6
Lung and bronchus	35	38.2	37.5	43.4	20	15.5
Male	42.3	44.9	51.3	52.3	25.9	21
Female	29.4	32.9	28	37	15.6	11.4
Prostate	18.9	17.9	37.9	225	87	15.4
Stomach	29	2.1	5	5.6	4.7	4.8
Male	3.8	2.9	7.7	7.7	6.0	5.9
Female	2.1	1.5	3.5	4.1	3.7	3.9
Uterine cervix	2.2	2	3.3	3.3	1.7	2.5

American Cancer Society. *Cancer Facts and Figures 2024*. Atlanta: American Cancer Society, 2024.

inactivity, and obesity. Additionally, enhancing early detection, such as mammography, colorectal screening, and Pap tests, as well as effective treatment and palliative care, can also contribute to reducing these disparities.¹⁷⁵

Disparities within rehabilitation care are also evident across various demographic factors, including age, sex, geographic location, SES, and insurance. Age-related disparities are notable, with older patients receiving less rehabilitation, with shorter inpatient rehabilitation admissions, despite their greater need.¹²⁵ Sex disparities are also significant. Research indicates that females are less likely to be referred for rehabilitation services compared to males. For instance, females are less likely to receive referrals for cardiac and stroke rehabilitation, resulting in less functional progress for these patients.^{52,92} Geographic disparities further complicate access to rehabilitation services. Patients in rural areas face challenges in securing outpatient therapies due to a lack of nearby facilities, leading to reduced or nonexistent rehabilitation postdischarge.^{57,113,150} Economic disparities also significantly hinder cancer rehabilitation efforts, as patients with lower SES often face marginally covered or uncovered costs for essential items, such as compression garments,

high copayments for physical and occupational therapy, and reduced home therapy benefits.^{27,144,146}

Health disparities are evident in both cancer care and rehabilitation care. Underrepresented minorities, individuals with lower SES, and those lacking insurance tend to experience higher rates of advanced disease and poorer outcomes in cancer treatment (see Table 30.2). Consequently, they may have an even greater need for rehabilitation services. However, their access to necessary rehabilitation care may also be hindered, compounding their overall health care challenges. Further study is needed to investigate the emerging disparities within cancer rehabilitation.²⁷

Staging

Cancer staging varies by cancer type but follows a general format to describe the spread of the disease from its origin. Commonly utilized is the T, N, and M system: T describes the characteristics of the primary tumor, N indicates the extent of regional lymph node spread, and M denotes the presence of distant metastases. Based on the TNM status, the disease is assigned a stage from I to

TABLE 30.3 Five-Year Survival Statistics for Different Cancers, 2013–2019

Cancer	FIVE-YEAR SURVIVAL (%)			Common Sites of Metastatic Spread
	Local	Regional	Distant	
Lung and bronchus	63	35	8	Brain, bone, liver, mediastinal lymph nodes
Breast	99	86	31	Brain, lung, bone, liver
Prostate	>99	>99	34	Bone, pelvic lymph nodes
Colon and rectum	91	73	14	Liver, lung
Ovary	92	73	32	Peritoneum, pleura
Uterine cervix	91	60	19	Peritoneum, lung, retroperitoneal lymph nodes
Uterine corpus	95	70	18	Retroperitoneal lymph nodes, lung
Pharynx and oral cavity	87	69	39	Lung, regional lymph nodes
Melanoma	>99	74	35	Brain
Stomach	75	35	7	Liver, lung, peritoneum
Esophagus	49	28	6	Liver, lung
Pancreas	44	16	3	Liver
Urinary bladder	71	39	8	Bone, intraperitoneal

American Cancer Society. *Cancer Facts and Figures 2024*. Atlanta: American Cancer Society, 2024.

IV. Stage I represents early, localized disease, and stage IV indicates advanced disease characterized by distant metastases, often considered incurable. Some cancers, such as primary brain tumors, are graded based on the degree of atypical cells observed histopathologically. Cancer may also be classified as in situ, local, regional, or distant, indicating whether it has remained in the layer of cells where it developed (in situ) or has spread beyond the tissue layer (local). The stage may direct treatment decisions and survival (Table 30.3). This information is also important for the physiatrist because it may influence selection of rehabilitation interventions and goals of care discussions with patients. Stage may also impact risk of recurrence or progression and guide additional interdisciplinary collaboration with the oncology care team when there is concern for potential progression of disease or recurrence.

Prognosis

The complexity of cancer presents a challenge to patients and clinicians because there are various treatment approaches, patterns of metastatic spread, and prognosis. Considerations may include likelihood of progression based on cancer type and stage, available treatment options, potential toxicities from these treatment options, and patient comorbid conditions. Additionally, treatment options are evolving as research continues to drive the field forward. Nonetheless, for the physiatrist and the individual patient, having some guidance regarding prognosis and expectations for disease course facilitates engagement in meaningful conversation regarding goals of care and treatment strategies. Table 30.3 shows the 5-year widely variable relative survival rates between 2013 and 2019 for various cancers, highlighting the implications of regional and distant spread at the time of diagnosis.⁴ Patients with prostate cancer have an excellent prognosis when detected early, but lung

cancer patients have a poorer prognosis, with a low 5-year survival rate. This information can inform rehabilitation goal setting, treatment approaches, and expectations. Understanding patterns of metastatic spread can also aid clinicians in directing their search for metastases because different cancers commonly spread to specific sites. For example, lung, breast, colon, and melanoma cancers commonly spread to the brain, highlighting the importance of neurologic screening exams; prostate, breast, lung, renal, and thyroid cancers commonly produce bone metastases. An online tool available for health care professionals to approximate survival is available at <https://cancersurvivalrates.com>. Questions to spur discussion among the interdisciplinary cancer and rehabilitation teams have been published: “Would you be surprised if this patient died in the next year?” or “Would you be surprised if this patient died in the next month?”¹⁵⁸ Additional predictors of survival include performance status, dyspnea, dysphagia, weight loss, xerostomia, anorexia, and cognitive impairment.¹⁷²

Stages of Cancer Rehabilitation

Dietz outlined four stages of rehabilitation goals: preventative, restorative, supportive, and palliative³⁹ (Table 30.4). This framework helps to establish meaningful goals, informed by cancer prognosis and treatment. They facilitate a person-centered rehabilitation treatment plan aimed at enhancing quality of life and potential future rehabilitation needs.

Cancer Treatments

Cancer treatments vary widely depending on numerous factors, including cancer disease type, stage, patient comorbidities, goals of care, and more. Treatment may include surgery, chemotherapy, radiation therapy, hormonal therapy, or targeted therapies potentially

TABLE 30.4 Dietz Stages of Cancer Rehabilitation

Preventive rehabilitation	Occurs prior to initiation of treatment. Goal is to help reduce functional decline from cancer, cancer treatment, and treatment side effects.
Restorative rehabilitation	Occurs after completion of treatment. Goal is to bring level of function back or close to pretreatment levels.
Supportive rehabilitation	Occurs during diagnosis of advanced cancer. Goal is trying to maintain current levels of function and prevent functional decline.
Palliative rehabilitation	Occurs during end of life. Goal is to maximize functional independence and autonomy, along with symptom control and caregiver/family training.

as a single modality or part of a multimodal treatment strategy. Each of these may have potential side effects (Table 30.5).

Surgery

Impairments resulting from surgery are influenced by the size, location, and type of the tumor. Healthy tissue may be impacted by surgical procedures aimed at local cancer control. The main reasons for removing normal tissue, despite the associated risk of

long-term negative effects, include accurate staging (such as sampling of lymph nodes or peritoneum), definitive removal of the tumor, ensuring local disease control (removal of lymph nodes that potentially contain cancer cells), and tissue harvesting for reconstruction. For the physiatrist, additional emphasis on the specific tissues and their potential functional ramifications is important, such as surgical interventions that may impact bone, nerve, muscle, lung parenchyma, and lymphatics. It is crucial to review patients' surgical reports to be aware of known sacrifice or potential compromise to the neuromuscular and lymphatic systems. Furthermore, normal postoperative healing may be impacted by administration of additional anticancer treatments, such as radiation therapy and/or chemotherapy.

There is a vast array of established surgical methods for tumor eradication. Operations that often require the attention of a physical medicine and rehabilitation specialist due to the expected sequelae impacting function include neck dissection for oropharyngeal carcinomas, limb salvage or amputation for osteosarcoma, resection of truncal or limb myosarcoma, axillary lymph node dissection, and pneumonectomy or lobectomy for lung neoplasms. Other procedures, such as nephrectomy, colectomy, mastectomy, and oophorectomy, may involve the resection of muscles, nerves, and/or vessels to achieve clean margins, which can cause acute functional losses. Physiatric involvement and rehabilitation care are crucial when there is surgical resection of central and/or peripheral nervous system tissues due to the potential for neuromuscular sequelae.

In addition to the primary impairments noted, secondary impairments can develop. These may result from altered biomechanics

TABLE 30.5 Treatment Modalities Utilized in Cancer Care and Their Potential Side Effects

	Surgery	Chemotherapy	Radiation Therapy	Hormonal Therapy	Targeted Therapies
Second cancers		×	×		
Fatigue	×	×	×	×	×
Pain	×	×	×	×	×
Cardiovascular changes: damage or increased cardiovascular disease risk		×	×	×	×
Pulmonary changes	×	×	×		
Neuropathy nerve injury	×	×	×		
Cognitive changes	×	×	×	×	×
Reproductive changes (e.g., infertility, early menopause, impaired sexual function)	×	×	×	×	×
Body weight changes (increases or decreases)	×	×		×	
Fat mass increases	×	×		×	
Lean mass losses	×	×		×	
Worsened bone health		×	×	×	
Musculoskeletal soft tissues: changes or damage	×		×	×	
Impaired immune function and/or anemia		×	×	×	×
Lymphedema	×		×		
Gastrointestinal system: changes or impaired function	×	×	×	×	×

Modified from Schmitz KH, Courneya KS, Matthews C, et al. American College of Sports Medicine Roundtable on Exercise Guidelines for Cancer Survivors. *Med Sci Sports Exerc.* 2010;42(7):1409.

or maladaptive compensatory strategies or movement patterns. Examples of this may include a secondary rotator cuff tendinopathy related to altered biomechanics from a spinal accessory neuropathy following a neck dissection in a head and neck cancer survivor or myofascial trapezius pain related to pectoralis muscle tightness following a mastectomy (+/- reconstruction) altering posture/positioning in a breast cancer survivor. These secondary impairments are amenable to physiatric intervention and rehabilitation care.

Chemotherapy

Chemotherapy is a common modality utilized in cancer treatment in both curative intent and palliative settings. Neoadjuvant therapy is used before surgery to reduce tumor size. Induction chemotherapy,

a term that at times is used interchangeably with neoadjuvant, refers to its use prior to radiation therapy. Chemotherapy may also be utilized concurrently with radiation therapy. Adjuvant chemotherapy is given after local control through surgery or radiation to eliminate undetectable micrometastases and reduce recurrence risk. Chemotherapy can also be utilized serially to manage incurable stage IV cancer temporarily.

A vast array of chemotherapeutic agents is used in oncologic practice. These drugs can be grouped into subclasses, including alkylating agents, platinum-based agents and their analogs, antimetabolites, topoisomerase inhibitor agents, antimicrotubule agents, differentiation agents, and miscellaneous agents. A nonexhaustive summary of these agents and their potential side effects are highlighted in Table 30.6. The type, dose, and duration of

TABLE 30.6 Classes of Antineoplastic Drugs

Class	Mechanism(s)	Commonly Used Agents ^a	Toxicities
Antitumor	Formation of covalent bonds of alkyl groups to DNA to form reactive intermediates that attack nucleophilic sites; the DNA can no longer function as a template	Mustards: chlorambucil, cyclophosphamide, ifosfamide Nitrosoureas: carmustine Tetrazines: dacarbazine Aziridines: mitomycin C, thiotepa Nonclassic alkylating agents: procarbazine	Myelosuppression (all), mucositis (busulfan), hepatotoxicity (busulfan, busulfan carmustine, dacarbazine), pulmonary fibrosis (busulfan, carmustine), cystitis (ifosfamide, cyclophosphamide), alopecia (cyclophosphamide), venoocclusive liver disease (busulfan, carmustine, mitomycin C)
Platinum based and their analogs	Platination of DNA with induction of apoptosis or arrest of cells in the G2 phase of the cell cycle; disruption of intracellular signaling pathways	Cisplatin, carboplatin, oxaliplatin	Nephrotoxicity, neuropathy, ototoxicity, myelosuppression
Antimetabolites	Interference with synthesis of DNA and RNA precursor molecules or DNA polymerase, thereby preventing DNA and RNA replication	Antifolates: methotrexate Fluoropyrimidines: fluorouracil Pyrimidine analogs: azacitidine Gemcitabine 6-thiopurines: 6-mercaptopurine, 6-thioguanine <i>Streptomyces parvulus</i> derivatives: actinomycin D	Nucleosides, azacitidine, gemcitabine 6-thiopurines), nephrotoxicity (methotrexate), neurotoxicity (methotrexate, arabinose nucleosides, azacitidine)
Topoisomerase-interactive agent	Interaction with DNA topoisomerases (enzymes regulating DNA packing, i.e., twisting and untwisting), leading to G2 phase arrest or apoptosis in S phase	Epipodophyllotoxins: etoposide, daunorubicin, epirubicin, idarubicin Camptothecin analogs: topotecan, irinotecan	Myelosuppression (all), mucositis (all), anthracyclines: doxorubicin, (anthracyclines), soft tissue ulceration postextravasation (anthracyclines), cardiotoxicity
Antimicrotubule agents	Disruption of microtubules that compose the mitotic spindle	Vinca alkaloids: vincristine, vinblastine, vinorelbine Taxanes: paclitaxel, docetaxel Miscellaneous antimicrotubule agents: estramustine	Peripheral neurotoxicity (vinca alkaloids, taxanes), gastrointestinal autonomic dysfunction (vinca alkaloids), neutropenia (vinca alkaloids), myelosuppression (taxanes), myalgias (taxanes), bradyarrhythmias (paclitaxel), fluid retention (docetaxel), skin toxicity (docetaxel), emesis (estramustine), congestive heart failure (estramustine)
Miscellaneous chemotherapeutic agents	Fludarabine: inhibits enzymes L-asparaginase: exploits tumor cells' inability to synthesize asparagine, limiting protein synthesis Bleomycin: free radical production of DNA breaks	Fludarabine, L-asparaginase, essential for DNA synthesis and repair	Myelosuppression (fludarabine), bleomycin immunosuppression (fludarabine), neurotoxicity (fludarabine), hypersensitivity reactions (L-asparaginase), pulmonary fibrosis (bleomycin), mucocutaneous toxicities (bleomycin)

^aLists not exhaustive.

chemotherapy vary across cancer types and stages, but common strategies involve combining agents from different classes to exploit complementary mechanisms of action, achieve synergy, and reduce normal tissue toxicity. Examples of common chemotherapy regimens include CHOP (cyclophosphamide, doxorubicin, vincristine, and prednisone), MOPP (mechlorethamine, vincristine, procarbazine, and prednisone), and AC-T (doxorubicin, cyclophosphamide, paclitaxel).¹⁶

Antineoplastic agents preferentially injure rapidly dividing cancer cells while relatively sparing normal cells. There can still be significant normal tissue toxicity, particularly in the skin and gastrointestinal systems. Common chemotherapeutic toxicities include peripheral neuropathy, cognitive dysfunction, cardiotoxicity, and pulmonary fibrosis. Proactive screening has reduced the incidence of significant cardiopulmonary toxicity. Risk factors for bleomycin-induced acute pneumonitis include cumulative dose, age, smoking, renal dysfunction, mediastinal radiation therapy, and oxygen administration. Doxorubicin-associated cardiac toxicity risk increases with cumulative doses.¹⁷ Chemotherapy-induced peripheral neuropathy (CIPN) is a prevalent complication, with drugs such as vinca alkaloids, cisplatin, ixabepilone, taxanes, and thalidomide causing peripheral neurotoxicity.¹⁴³ Symptoms, presentation, and severity vary and will be reviewed in more detail in the CIPN section.

Radiation Therapy

Approximately 50% of all cancer survivors will have received radiation therapy during their treatment course. Radiation therapy uses ionizing radiation, primarily in the form of photons or

particles to kill cancer cells. Photons, such as x-rays or gamma rays, or high-energy particles, such as protons, neutrons, and electrons, treat tumors by damaging the DNA of cancer cells and disrupting the cellular replication process with the goal of triggering cell death within the tumor. Radiation is delivered to the tumor via external beam radiation therapy outside the body or from within the body via brachytherapy.^{9,14,18} Unfortunately, the tissue in proximity of the tumor experiences both direct and indirect radiation injury. Like the tumor, healthy tissue within the radiation field receives direct injury causing DNA damage and triggering cell death. Indirect radiation injury is due to the formation of reactive oxygen and nitrogen species through the interaction of ionizing radiation with water molecules to form reactive oxygen species and with endogenous nitric oxide to form reactive nitrogen species. These molecular moieties injure healthy cells, triggering an inflammatory response, leading to recruitment of fibroblasts, which then deposit extracellular matrix components into the surrounding tissue, resulting in radiation-induced fibrosis (RIF).^{154,164,167} (Fig. 30.2). These radiation effects are classified as acute (occurring during or immediately after completion of treatment), early delayed (occurring ≤ 3 months following completion of radiation treatment), and late delayed (occurring > 3 months following the completion of radiation treatment). Early effects are usually temporary and affect rapidly proliferating tissues such as skin and mucosa. Late effects are often irreversible and impact tissues with low proliferative potential, such as connective tissue or neural tissue, with mechanisms involving microvascular damage, cellular apoptosis, and endothelial changes.¹¹⁹

Radiation fibrosis syndrome (RFS) is typically a late complication of radiation therapy, often manifesting years after treatment

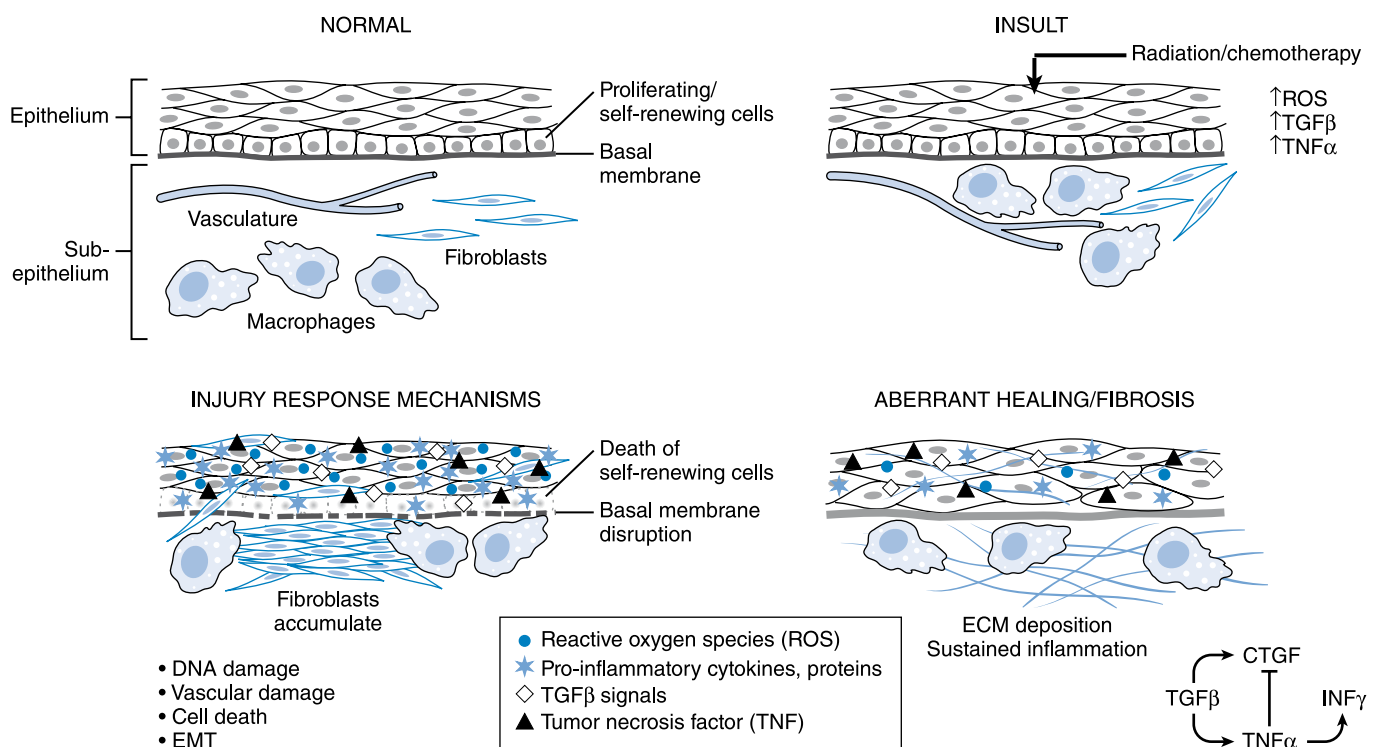


Fig. 30.2 Sequential pathogenesis of radiation fibrosis. CTGF, Connective tissue growth factor; DNA, deoxyribonucleic acid; ECM, extracellular matrix; EMT, epithelial to mesenchymal transition; INF γ , interferon gamma; ROS, reactive oxygen species; TGF α , transforming growth factor alpha; TGF β , transforming growth factor beta; TNF, tumor necrosis factor.

and progressing unpredictably. It is a chronically progressive pathologic process with symptoms that change over time due to the deterioration of affected structures. This sclerosing disease process can adversely affect any tissue type, including skin, musculoskeletal, nerve, and viscera, leading to a constellation of neuromuscular, musculoskeletal, vascular, and/or organ system, resulting in related clinical manifestations and complications defined as RFS (Table 30.7). The symptoms patients may experience will depend heavily on the radiosensitivity of the tissues, organs, and structures exposed in the radiation field, radiation total dose, dose per fraction, treated tissue volume, and treatment duration. Additionally, concurrent chemotherapy, prior or subsequent surgery, preexisting connective tissue diseases, and certain genetic predispositions contribute to a higher likelihood of developing RFS.¹⁵⁵

RIF can cause peripheral or central nervous system (CNS) dysfunction through fibrotic- and stenotic-induced ischemic changes within the vaso vasorum and compressive and traction forces of soft tissues surrounding the nerves. These changes cause demyelination and axonal loss.^{62,108} This is commonly seen in conditions such as radiation-induced cervical and brachial plexopathy, a common side effect of radiation treatment in patients with a history of head and neck cancer. Additionally, these neurologic changes may also present as autonomic dysfunction (e.g., orthostatic hypotension, baroreflex failure, sexual dysfunction, bladder dysfunction, and arrhythmias).^{124,156}

Neuropathic pain, which arises from abnormal ectopic activity in neural structures that transmit pain signals to the spinal cord, thalamus, and brain, is common in RFS. It is typically out of proportion to the perceived pathology. RIF can lead to the formation

of myopathic muscles. Patients typically experience weakness, intermittent spasms, and pain, which is often exacerbated by radiation-induced neurologic dysfunction. The progressive fibrotic process also results in a loss of elasticity in tendons and ligaments, thus adversely affecting mobility of joints involved in the field of radiation. In many cases, joint injury may be indirect due to neuromuscular changes rather than direct injury to the joint. Most patients experience acute radiation-induced dermatitis, which initially improves but is later followed by chronic radiation-induced dermatitis. Symptoms include poor wound healing, changes in skin appearance and texture, necrosis, and ulcerations. Radiation effects on bone can result in accelerated osteoporosis, development of osteonecrosis, abnormal bone growth, and decrease in hematopoietic function.^{78,156} RIF is a significant contributor to the development of lymphedema, especially in the setting of lymph node dissection. This can affect any part of the body where lymph nodes have been exposed, such as the neck, breast, trunk, pelvic region, and upper and lower extremities. RFS can cause microvascular injury and can lead to accelerated atherosclerosis, obliterative endarteritis, peripheral vascular disease, venoocclusive disease, thromboembolism, and venous insufficiency. These vascular changes also result in worsening ischemic changes to the tissue and organs supplied by the affected blood vessels.^{122,184}

Many patients experience neuromuscular system, autonomic system, and organ system dysfunction as a result of RFS (e.g., Hodgkin lymphoma patients typically experience endocrine dysfunction [hypothyroidism], neuromuscular dysfunction [cervicalgia, cervical plexopathy, brachial plexopathy, scapular winging], musculoskeletal dysfunction [dropped head syndrome, cervicalgia], pulmonary dysfunction [radiation-induced pulmonary fibrosis of the lungs], cardiovascular dysfunction [baroreceptor reflex failure, cardiac valvular dysfunction, cardiac fibrosis], and gastrointestinal dysfunction [dysphagia] due to mantle field radiation treatment). RFS symptoms in breast cancer patients typically include lymphedema, adhesive capsulitis, and brachial plexopathy. To minimize adverse radiation effects, radiation oncologists attempt to focus the radiation on tumors while protecting surrounding healthy tissues.

Radiation oncologists must carefully evaluate multiple factors when planning treatment, including tissue radiosensitivity, irradiated volume, radiation dose, and fractionation schedule. Radiation dose is measured in grays, with patients feeling no pain during treatment due to the nature of ionizing energy and lack of heat generation. To minimize side effects on healthy cells, radiation oncologists utilize a variety of dose delivery techniques to concentrate the radiation administered on tumors while limiting exposure to surrounding tissues. When developing a treatment plan, radiation oncologists also consider whether treatment is intended to be curative or palliative, and the patient's ability to tolerate radiation treatment, because this influences the acceptable level of toxicity. This comprehensive assessment is crucial for balancing potential toxicities against the likelihood of therapeutic benefits and long-term quality of life for each patient.^{10,60,70}

Hormone or Endocrine Therapies

Hormone or endocrine therapies are systemic treatments utilized to treat malignancies whose growth is promoted by hormones. These are most commonly employed as part of the treatment for breast and prostate cancer, but they have additional indications (endometrial, ovarian, and adrenal cancers).

Tamoxifen is a selective estrogen receptor modulator that is utilized in the treatment of hormone receptor positive breast cancer,

TABLE 30.7 Late Effects of Radiation Fibrosis

System	Adverse Late Effects
Endocrine	Hypothyroidism, hypogonadism, adrenal insufficiency, glucose intolerance due to pancreatic insufficiency
Exocrine	Xerostomia, pancreatic enzyme deficiency
Neural	Myelopathy, plexopathy, cerebrovascular ischemia, dementia, leukoencephalopathy, cranial neuropathy
Lymphatic	Lymph node necrosis, lymphedema
Gastrointestinal	Dysmotility, malabsorption, neuroconstipation, obstruction, perforation, dysgeusia
Dermis	Atrophy, ulceration, delayed healing, hyperpigmentation
Auditory	Progressive loss of acuity, tinnitus
Vascular	Premature atherosclerosis, venous sclerosis
Pulmonary or upper respiratory	Parenchymal fibrosis tracheal stenosis, dysphonia secondary to laryngeal fibrosis
Musculoskeletal	Fibrosis, osteonecrosis, osteoporosis, soft tissue necrosis joint contracture,
Ocular	Corneal ulceration, retinopathy, scleral necrosis
Genitourinary	Neurogenic bladder, renal failure, obstruction, perforation

breast cancer risk reduction in high-risk populations, and ovarian cancer.^{13,101} It functions by competitively binding to estrogen receptors on tumor cells and other organ tissues, acting as an anti-estrogen in the breast tissue but a weak estrogen agonist in the endometrium. Notable risk factors associated with tamoxifen include an increased risk of stroke, pulmonary embolism, endometrial adenocarcinoma, and uterine sarcoma.⁴⁹ The impact of tamoxifen on bone varies based on menstrual status. It is associated with protective benefits on bone mineral density in postmenopausal females but a decline in bone mineral density in premenopausal females. Side effects include musculoskeletal pain with arthropathy and myalgia, although at lower rates than aromatase inhibitors.¹⁰⁶

Aromatase inhibitors (AIs) inhibit the enzyme aromatase, which decreases plasma estrogen levels (estrone, estradiol, estrone sulfate). Medications in use include anastrozole, exemestane, and letrozole. These are commonly prescribed as adjuvant endocrine therapy for postmenopausal hormone receptor-positive breast cancer or premenopausal breast cancer in conjunction with ovarian function suppression. Additionally, some are utilized for recurrent ovarian cancer.^{36,127} Common side effects include osteoporosis, increased fracture risk, increased incidence of ischemic heart disease, weight gain, edema, nausea, fatigue, depression, hot flashes, and night sweats. Of particular note for rehabilitation providers is the high frequency of musculoskeletal side effects, known as aromatase inhibitor-induced musculoskeletal symptoms (AIMSS). AIMSS is a common sequelae impacting 47% to 82%, which can lead to discontinuation of therapy in nearly one-fourth of cases.^{12,32,95} Therefore it is crucial to address these symptoms to ensure that patients can continue their treatment. The approach to managing AIMSS varies depending on whether the symptoms present more focally as arthralgias or tendinopathies or more generalized diffuse pain. Focal symptoms can be treated based on the rehabilitation strategies utilized for their etiologies unrelated to AIs. Diffuse symptoms require care for the entire body. Participating in 150 minutes of aerobic exercise per week and strength training has been shown to decrease pain.⁷⁴ Yoga and acupuncture were found to improve symptoms in small studies.^{31,54} Supplements such as glucosamine (1500 mg/day) with chondroitin sulfate (1200 mg/day) has shown moderate pain improvement.⁵⁸ Vitamin D has mixed results in studies.^{128,138} Pharmacologic options include nonsteroidal antiinflammatory drugs (NSAIDs) and acetaminophen used empirically.⁸³ Duloxetine was shown to have improvement in pain related to AIMSS in a randomized controlled trial.⁶⁷

Luteinizing hormone-releasing hormone (LHRH) agonists, also known as LHRH analogs or gonadotropin-releasing hormone agonists, are medications that reduce the production of testosterone. These include goserelin, leuprolide, and triptorelin.¹¹⁵ Antiandrogen medications or androgen deprivation therapies include apalutamide, enzalutamide, darolutamide, bicalutamide, flutamide, and nilutamide.¹¹⁵ These medications bind to androgen receptors, preventing androgens from promoting tumor growth. Side effects may include osteoporosis, loss of muscle mass, weight gain, and fatigue.

Targeted Therapies

Targeted therapies are therapeutic agents that are specifically designed to target small molecules responsible for the proliferation of cancer. There are several types of targeted therapies, which include angiogenesis inhibitors (bevacizumab), monoclonal antibodies (trastuzumab, cetuximab), proteasome inhibitors (bortezomib), and signal transduction inhibitors (imatinib). Please note this

section only provides a brief overview and is not an all-encompassing review.

Trastuzumab, a recombinant humanized monoclonal antibody, is utilized in the treatment of HER2-positive breast, uterine, gastric, colorectal, and esophageal cancers. It downregulates proliferation of cells, which overexpress HER2. A notable potential adverse reaction is cardiomyopathy, decreased left ventricular ejection fraction, and arrhythmias. The cardiomyopathy is more prevalent in patients who also received an anthracycline.⁷⁹

Bevacizumab, a recombinant humanized monoclonal antibody, inhibits the activity of vascular endothelial growth factor, a proangiogenic growth factor. It is utilized in treating metastatic colorectal cancer, non-small cell lung cancer (NSCLC), and breast and ovarian cancers. However, cardiotoxic effects, such as angina, heart failure, hypertension, stroke, myocardial infarction, and arterial thromboembolic events, are potential side effects. The cardiovascular risk is increased when used in conjunction with anthracycline. Patients older than 65 years of age or those with previous arterial thromboembolic events face a higher risk.^{51,107}

Tyrosine kinase inhibitors, including imatinib, nilotinib, dasatinib, and ponatinib, are commonly used and effective in the treatment of chronic myeloid leukemia. Potential side effects include fatigue, weight change, gastrointestinal symptoms, rash, cardiovascular disease, pulmonary disease, cytopenias, cognitive impairment, peripheral neuropathy, arthralgias, and myalgias.¹⁵³

Immunotherapy

The utilization of immunotherapy has been advancing recently with the development of monoclonal antibodies to leverage the regulatory molecules of the immune response—immune checkpoint inhibitors. These novel medications include ipilimumab, pembrolizumab, nivolumab, and durvalumab. These can be used for advanced melanoma, renal cell carcinoma, NSCLC, bladder cancer, and lymphomas. Ipilimumab inhibits cytotoxic T-lymphocyte antigen-4-enhancing immune response, enabling cytolytic T cells to identify and trigger apoptosis. Pembrolizumab and nivolumab are monoclonal antibodies that bind to the programmed cell death-1 (PD-1) receptor to block the binding of its ligands, PDL-1 and PDL-2. By blocking PD-1, these treatments undermine the immune resistance strategies utilized by cancer cells. While these innovative immunotherapies are generally well tolerated, they can cause immune-related adverse events, including colitis/diarrhea, rash or dermatitis, hepatitis, pneumonitis, and endocrinopathies. There are emerging data regarding immune-related neuromuscular events, which may be experienced by 1% to 3% of patients and include myasthenia gravis, myositis, acute inflammatory demyelinating polyneuropathy, and chronic inflammatory demyelinating polyneuropathy.^{86,111}

Common Conditions Related to Cancer and Cancer Treatments

Fatigue

Cancer-related fatigue is a common and distressing symptom affecting patients at all stages of their disease, from initial presentation to survivorship, and can persist for months to years. It negatively impacts function and quality of life, with its prevalence reported to be as high as 70% to 99%.^{43,56} However, this symptom is often underrecognized and undertreated, with historically 50% of patients not discussing treatment options for fatigue with

their oncology care team, despite describing it as impacting their lives more than pain.¹⁷³ The National Comprehensive Cancer Network (NCCN) defines cancer-related fatigue as “a distressing, persistent, subjective sense of physical, emotional, and/or cognitive tiredness or exhaustion related to cancer or cancer treatment that is not proportional to recent activity and interferes with usual functioning.”⁸ Patients may experience difficulty concentrating, emotional exhaustion, unrefreshing sleep, insomnia, or hypersomnia.⁴³ It is crucial to acknowledge and treat fatigue to reduce symptom burden and improve patients’ function and quality of life.

As part of the management strategy, it is important to evaluate for and treat reversible sources of fatigue through laboratory testing or sleep studies. Potential reversible sources are listed in [Box 30.1](#), and considerations for laboratory analysis are provided in [Table 30.8](#). Cancer-related fatigue can occur without identified etiology and may persist for years. The NCCN published guidelines for providers highlighting the importance of screening, evaluation, and treatment strategies, and they have also published a resource for patients.¹¹⁶ Potential cooccurring deconditioning and mood disorders should also be addressed. There is evidence supporting recommendations for physical activity, yoga, massage therapy, cognitive behavioral therapy, and psychoeducational therapies.⁷⁶ Psychostimulants may also be considered, but further research is required.^{76,109} For patients at the end of life, short-term use of corticosteroids has support to reduce fatigue, but long-term use is limited due side effects.¹⁸³

• BOX 30.1 Reversible Sources of Cancer Fatigue

Anemia
Lack of restorative sleep or sleep-related problem
Cytokine release (e.g., tumor necrosis factor)
Hypothyroidism
Hypogonadism
Depression
Deconditioning
Steroid myopathy
Centrally acting medications
Pain
Adrenal insufficiency
Cachexia

TABLE 30.8 Laboratory Testing to Evaluate Fatigue

Electrolytes	Sodium, potassium, chloride, bicarbonate
Chemistry panel	Creatinine, blood urea nitrogen, glucose, magnesium, calcium, phosphorus, total bilirubin, serum transaminases, alkaline phosphatase, lactic dehydrogenase, albumin, total protein
Endocrine panel	Thyroid-stimulating hormone, T4, serum testosterone
Anemia panel	Complete blood count, iron, total iron binding capacity, ferritin

Cognitive Impairment

Cancer-related cognitive impairment (CRCI) is common during active cancer treatment, impacting up to 75% of non-CNS cancers and 90% of CNS cancers.³ Of cancer survivors, 30% to 60% continue to experience cognitive symptoms even after treatment ends.^{1,3,170} Although it is often colloquially referred to as “chemobrain,” it is not as accurate as CRCI because chemotherapy is not the single contributor to the etiology of these symptoms; it can be present in cancer survivors who have not been treated with chemotherapeutic agents.¹⁰⁵ Individuals may experience difficulty with executive function, concentration, memory, processing speed, and short-term memory. This can be a major source of distress with deleterious impacts on school/work performance, social functioning, and community reintegration.^{1,129,171} Several potential mechanisms may contribute to CRCI, including impaired blood-brain barrier integrity, cell death in the hippocampus and corpus collosum, genetic variability or polymorphisms, DNA damage, cytokine production, telomere shortening, defects in neural repair, hormone dysregulation, and neurotransmitter defects.^{19,38,48,75} Whole-brain radiation is also known to be neurotoxic.¹⁹ Additional influences may include chronic stress, loneliness, medical comorbidities, hormonal changes, physical inactivity, depression, anxiety, insomnia, and side effects from other medications.^{6,66,136}

Evidence guiding management of CRCI is limited. Thus far no brief screening tool has been identified to be effective.¹¹⁷ Neuropsychological testing helps assess alterations to specific functions; however, this requires specialized providers and significant time, limiting the ability to scale its use more broadly.¹³⁵ It may be appropriate to pursue neuropsychological testing in some instances, such as cancer survivors with symptoms persisting for several months.¹¹⁷ It is important to assess for potential reversible factors contributing to CRCI ([Box 30.2](#)). Neuroimaging is utilized for patients presenting with focal neurologic deficits, especially for those with known CNS metastatic disease or for those who are at high risk for such metastases, to rule out structural abnormalities. It is important to provide patients with validation of their experience with CRCI by providing appropriate reassurance and support with self-management and coping strategies. First-line interventions include cognitive rehabilitation, psychotherapy, and physical activity. Additional nonpharmacologic recommendations include meditation, yoga, mindfulness-based stress reduction, and cognitive training. Although evidence is limited, second-line treatment medications (methylphenidate, modafinil, donepezil) may be considered in selective cases.¹¹⁷

Lymphedema

Lymphedema is a chronic regional accumulation of protein-rich fluid.¹⁷⁶ Cancer and its treatment can lead to the disruption of the

• BOX 30.2 Reversible Sources of Cancer-Related Cognitive Impairment

Depression and anxiety
Sleep-related problem
Pain
Medications
Vitamin or hormonal deficiencies
Delirium

lymphatic system leading to this secondary lymphedema. Risk factors include higher number of lymph nodes surgically removed, radiation therapy field (including nodal region), and obesity.⁸⁵ Lymphedema presents with localized swelling, depending on the location of lymphatic system disruption, such as arm swelling in breast cancer survivors. Head and neck cancer survivors may experience swelling externally with visible changes in the face and neck or internally impacting the aerodigestive tract. Of breast cancer survivors, 20% to 30% develop lymphedema.^{34,145} For head and neck cancer survivors, newer studies have found a higher incidence up to 90% when evaluating for both internal and external lymphedema.¹³² This localized swelling can detrimentally impact physical symptoms, functional limitations, and quality of life.^{37,65,68} If lymphedema severity progresses, patients may develop skin changes, including hyperpigmentation, hyperkeratosis, papillomatosis, and lymphorrhea. Before delving into lymphedema, it is important to note that not all limb swelling in cancer survivors is lymphedema. Box 30.3 lists potential causes of swelling in cancer survivors.

Lymphedema is categorized into stages, from 0 to 3. Stage 0 or 1a, also known as the latency or subclinical stage, is when a patient is at risk of developing lymphedema due to underlying impaired lymph transport; however, there is no visible swelling at this stage. There may be subtle changes in tissues or subjective

• BOX 30.3 Other Potential Causes of Limb Swelling in Cancer Survivors

Venous thrombosis
Heart failure
Hepatic disease
Renal disease
Thyroid dysfunction
Hypoproteinemia
Medications (corticosteroids, calcium channel blockers, gabapentin, taxanes)
Lipedema

symptoms. Stage 1 is the spontaneous/reversible stage, characterized by reversible swelling that improves with elevation. The edema is soft and pitting. Stage 2 is spontaneously irreversible with harder swelling with increased limb volume that does not improve with elevation. Volume reductions are still possible with appropriate treatment. This is when patients begin to experience tissue fibrosis and skin thickening. Stage 3 is marked by skin changes (hyperpigmentation, hyperkeratosis, papillomatosis) and fibrosis, which is also referred to as lymphostatic elephantiasis⁴⁶ (Fig. 30.3).

Early detection and intervention are essential. In addition to visible changes, lymphedema can also contribute to functional limitations due to pain, discomfort, and altered biomechanics. Individuals with lymphedema are at increased risk of developing cellulitis. In addition to physical symptoms, patients may experience psychosocial effects as changes in appearance can impact their quality of life and lead to higher levels of distress, anxiety, and depression.⁷ Appropriate treatment of lymphedema can decrease swelling, lower cellulitis rates, and decrease hospitalizations.¹⁷⁸

Evaluation of lymphedema may involve imaging and volume measurements. Lymphoscintigraphy, which involves injecting a radiolabeled tracer, aids in visualizing lymphatic dysfunction. Near-infrared fluorescence imaging with indocyanine green lymphography is also used to identify lymphatic dysfunction. To monitor progression, several methods are routinely used for volume measurements, including circumferential tape measurements and perometry. Bioimpedance spectroscopy is increasingly being utilized to assess for subclinical lymphedema.

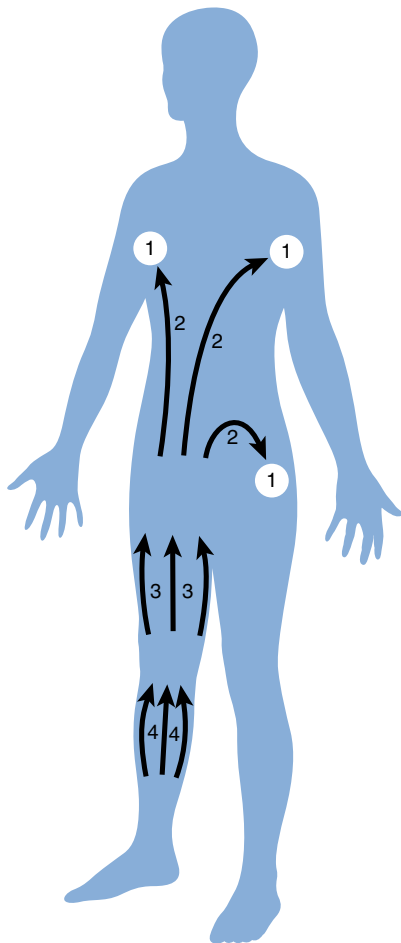
There is growing evidence to support a prospective surveillance model to reduce progression to and rates of chronic breast cancer-related lymphedema, improving patient outcomes and reducing health care costs.^{102,130,152} Recently, the NCCN survivorship guideline also added a specific inclusion of the use of bioimpedance spectroscopy when available.¹³⁷ In this model, routine surveillance is conducted every 3 months for the first 3 years and every 6 months in years 4 and 5. If subclinical lymphedema is identified, patients initiate treatment with a compression garment of 23 to 32 mm Hg daily for 4 weeks. Should the patient exhibit



• **Fig. 30.3** Lower extremity lymphedema before (A) and after (B) complete decongestive therapy, which afforded dramatic volume reduction.

any observable swelling or undergo a change in limb size based on circumference measurements, alongside the use of compression garments, the patient should be directed to a certified lymphedema therapist for complete decongestive therapy (CDT).¹⁰²

Standard of care for lymphedema management is CDT. Lymphedema therapy consists of two stages. The first component of CDT under guidance of a lymphedema therapist includes manual lymphatic drainage (MLD), short-stretch compressive bandaging, skin care, therapeutic exercise, and compression garments. The goal of the initial phase is to decrease lymphedema volume. This phase involves regular sessions of MLD with a lymphedema therapist, followed by compression bandaging and exercises. After maximum volume reduction is achieved, patients initiate a long-term maintenance program. In this phase, compressive garments are worn during the day. MLD, also referred to as lymphatic massage, is a specialized technique to stimulate lymph vessels, leading to increased transport of the congested lymphatic system. MLD treatments start in lymphostatic regions adjacent to functional lymphotomes and gradually progress to distal regions (Fig. 30.4). Skin care is emphasized to maintain skin



• **Fig. 30.4** Manual lymphatic drainage sequence in the treatment of right lower limb lymphedema due to inguinal lymph node dissection: 1, Stimulate intact lymph node beds where the stagnant lymph will be directed; 2, clear the pathways that will be used to redirect stagnant lymph into functioning lymphotomes; 3, direct stagnant lymph proximally along the cleared pathways, working backward into the congested territory; 4, complete treatment with proximal redirection of lymph from the most distal portions of the lymphedematous territory.

integrity, prevent microfissuring, decrease overgrowth in skin crevices, and prevent infection.¹⁰²

Microvascular surgical techniques, such as lymphovenous bypass or lymphovenous anastomosis and vascularized lymph node transfer, to improve residual lymphatic function are becoming more prevalent. Lymphovenous bypass is a procedure that establishes connections between lymphatic and venous microvessels. Vascularized lymph node transfer involves removing a lymph node or nodes from the omentum or an intact node bed and relocating it to the congested area. Although growing and evolving, the overall level of supportive evidence is still limited, but there may be potential utility in such procedures as a preventative intervention at the time of axillary lymph node dissection (ALND). Based on the safety profile and feasibility of lymphovenous anastomosis, preliminary reports suggest that preventative lymphatic surgery may help prevent chronic cancer-related lymphedema. Nonetheless, a systematic review highlighted the lack of high-quality studies on preventative lymphatic surgery, limiting recommendations for this approach.²⁶

Peripheral Neuropathy

CIPN is a condition resulting from damage to peripheral nerves. It can cause symptoms such as numbness, tingling, pain, weakness, and gait abnormalities due to the involvement of motor and/or sensory nerves. Autonomic neuropathy is less common but may present with orthostatic hypotension, constipation, urinary difficulties, or sexual dysfunction. These symptoms can interfere with daily activities and reduce quality of life. They may also limit the amount of chemotherapy a patient can tolerate. The incidence of this condition is dependent on several factors, including the chemotherapy drug(s) used, dosage, patient's comorbidities, and method of assessment. Drugs associated with neuropathy include taxanes, vinca alkaloids, platinum compounds, bortezomib, and thalidomide. These medications can cause different symptoms based on their mechanisms. For instance, vinca alkaloids and taxanes inhibit microtubules leading to axonal damage, presenting with a length-dependent distal symmetric motor and sensory neuropathy.⁹⁰ Symptoms may appear early in treatment and progress with dose and duration, typically improving after treatment completion.¹⁶³ In contrast, platinum compounds cause predominantly sensory symptoms by accumulating in the dorsal root ganglia and causing DNA damage resulting in symptoms that may present and worsen for months after treatment, known as a coasting phenomenon.^{126,140} Of note, patients with Charcot-Marie-Tooth disease require special consideration because they are at risk of severe neuropathy following treatment with vincristine.⁶⁹

As part of evaluation and treatment of CIPN, a thorough patient history is vital, including details on symptoms, onset, timing in relation to treatment, distribution, severity, pain assessment, functional assessment, and the patient's ability to carry out daily activities. Lab tests can help exclude other possible causes of neuropathy, such as paraneoplastic neurologic disorders or metabolic/nutritional deficiencies. Refer to Box 30.4 for a broad but not all-encompassing list of potential etiologies for neuropathic symptoms. There are currently no approved treatments that effectively prevent CIPN. Regular monitoring during chemotherapy is recommended due to the potential for symptom progression.⁸² Pain and paresthesia management strategies are often based on treatments proven effective for painful diabetic neuropathy, with duloxetine and pregabalin showing efficacy in studies specific to CIPN.^{120,134,148} Other potential pharmacologic options include gabapentin, venlafaxine, amitriptyline,

• BOX 30.4 Potential Etiologies for Neuropathic Symptoms in Cancer Survivors

Alcohol
Amyloid
Central nervous system lesions
Complex regional pain syndromes
Diabetes mellitus
Entrapment syndromes
HIV neuropathy
Hypothyroidism
Hereditary sensory neuropathies
Multiple sclerosis
Paraneoplastic syndromes
Plasmacytoma
Postherpetic neuralgia
Radiation fibrosis related injury (plexopathy; myeloradiculoplexoneuromyopathy)
Vitamin deficiencies (B and E vitamins, folate, copper)

valproate, and capsaicin.⁸² In small samples, electrical stimulation and acupuncture showed benefit.^{149,181} While pharmacologic interventions may not alleviate the symptoms of numbness and weakness, rehabilitation specialists have a critical role to help manage these. Interventions include patient education on fall prevention, therapeutic exercises, progressive gait and balance training, home modifications, assistive devices, and orthotics.

Pain

Pain is common for cancer survivors, impacting 66% of patients with advanced disease and 39% following curative intent treatment.¹⁶⁹ The etiology of this pain may be due to the direct involvement of the tumor, treatment effects, or factors unrelated to the cancer or its treatment, such as comorbid musculoskeletal conditions.^{20,84} Severe pain can negatively affect functionality and quality of life. Given that the cause of pain during cancer treatment and survivorship can be due to various factors, it is crucial to identify the pain generator(s) to guide treatment approaches effectively. This can help reduce pain and optimize function and quality of life. Pain management strategies can include targeted therapy referrals, therapeutic exercise, modalities, oral and topical medications, and injection procedures. Studies have highlighted the positive impact of patient education on managing cancer pain. A review article emphasized key principles, including pain education, patient-centered assessment and management, the importance of patient empowerment, and the integral role of patient education in routine care.⁹⁷ A meta-analysis also found interventions that fostered “enablement” to be effective.¹⁰⁰ Pain may be intertwined with fear of cancer progression or recurrence. Therefore education about pain generators, pathophysiology, and treatment plans may help alleviate fear and enhance a sense of control. By providing education, there is enhanced potential for shared decision making within the range of treatment options.

Physical therapy and therapeutic exercise can help stabilize painful areas and alleviate myofascial pain.²¹ A systematic review focusing on breast cancer survivors showed that physical therapy was effective in reducing pain and enhancing range of motion (ROM).⁵⁹ Isometric contractions that avoid pain-causing movements may be better tolerated.²³ A systematic review found that exercise is beneficial at all stages of cancer treatment, for all types of cancer, and can enhance physical function and alleviate impairments.¹⁵¹ Important

safety considerations and precautions are essential to account for coexisting conditions, such as bone metastases, peripheral neuropathy, or cardiopulmonary dysfunction. This is reviewed later in this chapter.

Modalities may also be utilized to help ameliorate pain. These may include topical heat, cryotherapy, and transcutaneous electrical nerve stimulation. Complementary and alternative medicine techniques, including acupuncture, cupping, music therapy, and reflexology, may also help alleviate cancer pain. However, more research is required to substantiate their benefits. The use of braces, orthotics, assistive devices, and compensatory strategies can also be beneficial in reducing pain and enhancing function.²³ Osteopathic manipulative treatment (OMT) outside the cancer population has shown benefit in addressing pain, and an emerging body of evidence also indicates that OMT may help lessen pain and enhance function for cancer survivors. With appropriate safety precautions, most OMT techniques may be used for cancer survivors, but more research is needed.¹⁰³

There are numerous tools available to enhance physical mobility and function, including orthotics, braces, and assistive devices such as sock aids and long-handled reachers. For cancer survivors whose functional capacity is limited, these tools can be crucial in maintaining independence, which can in turn impact their mental health and quality of life.²¹ The provision of assistive devices and bracing depends on the individual patient. Offloading of painful joints such as with the use of a cane or walker can decrease pain. Bracing can also help optimize posture and alignment, such as using a Headmaster or Velpeau collar as needed for individuals with dropped head syndrome.⁸¹

There are many medications available to help ameliorate pain. These may include analgesics such as acetaminophen, NSAIDs, or opioids. The choice of medication depends on the pain generator. For instance, when pain is due to bone metastasis, NSAIDs, bisphosphonates, or radiotherapy may be helpful.² Other frequently used medications for neuropathic pain include gabapentin, pregabalin, duloxetine, venlafaxine, amitriptyline, nortriptyline, and topical capsaicin.^{82,153}

Injection procedures may also be appropriate as part of the comprehensive approach to pain for cancer survivors. This may include corticosteroids, local anesthetics, and botulinum toxins. For myofascial pain, dry needling and trigger points may be utilized.²¹ Corticosteroids can be utilized in a similar fashion as practiced in general rehabilitation but may require precautions with the oncology care team during active treatment particularly in cytopenic patients. Botulinum toxin injections can be utilized as part of the management of cervical dystonia related to late effects of radiation fibrosis or trismus as well as for specific etiologies of postmastectomy pain.^{89,162} Additional injections utilized for postmastectomy pain include serratus plane blocks and intercostobrachial nerve blocks.^{179,185} These are just a few examples of the vast cadre of injection techniques that may be utilized.

Rehabilitation of Specific Cancer Populations

Breast Cancer

Breast cancer survivors are at high risk of sequelae, with seven of eight experiencing upper extremity symptoms.¹⁰⁴ Neuromuscular, musculoskeletal, and functional impairments may include shoulder dysfunction, post-breast surgery pain syndrome, AIMSS, and lymphedema. They may also present with brachial plexopathies, radiculopathies, and mononeuropathies. Fatigue, cognitive

impairment, and CIPN are also prevalent among this group and were discussed earlier in this chapter. Postmastectomy pain syndrome, also known as post-breast surgery pain syndrome, is a chronic pain condition that can occur after any breast surgery.¹⁷⁴ Its prevalence ranges widely, but may be up to 68%.^{77,98} This syndrome encompasses a variety of diagnostic subgroups that can stem from different causes, including shoulder dysfunction, chest wall pain, radiation fibrosis, axillary web syndrome, intercostobrachial neuralgia, and phantom breast pain. Identifying the specific cause of pain is crucial to inform the appropriate treatment, which can include physical therapy, pharmacotherapy, and injection procedures. Physical therapy is a cornerstone of management and may include therapeutic exercise, neuromuscular reeducation, progressive resistance training, desensitization techniques, and myofascial release. Medication options include NSAIDs, gabapentin, pregabalin, and duloxetine depending on the pain generator (e.g., a short course of NSAIDs for pain related to a rotator cuff tendinopathy or gabapentin for neuropathic pain).^{5,88} Depending on the cause of the pain, procedures may include intercostobrachial nerve block, serratus plane block, or botulinum toxin injections.^{88,179}

Breast cancer survivors can experience shoulder dysfunction, characterized by pain, restricted ROM, stiffness, weakness, and impaired function, often due to rotator cuff pathology and adhesive capsulitis. This is often grouped together as post-breast surgery pain but highlights the importance of identifying the pain generator to tailor treatment appropriately. Not only does shoulder morbidity affect the shoulder ipsilateral to the breast cancer, but the contralateral shoulder can also be negatively impacted, exhibiting altered kinematics and muscle activity compared to healthy individuals. Surgery and radiation fibrosis can impact shoulder positioning and function due to guarding of the shoulder movements after treatment as well as pectoralis muscle hypertonicity and shortening contributing to altered biomechanics.^{24,42} Additional factors may include axillary web syndrome (Fig. 30.5) and lymphedema.^{68,166} Physical therapy is a crucial part of treatment, with a Cochrane review recommending prompt postoperative implementation due to significant improvements in shoulder ROM.^{161,163}



• **Fig. 30.5** Axillary web syndrome manifest by thick, fibrous cords tethering the arm.

NSAIDs can help manage mild to moderate pain, and corticosteroid injections may provide temporary relief.^{24,159}

AIMSS may affect as many as 82% of females taking AIs for hormone-sensitive breast cancer and can lead to discontinuation of therapy in 24% of cases.^{32,93,95} Therefore addressing these musculoskeletal complaints is crucial to help support continued use of AIs. Treatment approaches vary based on whether the symptoms are focal arthralgias or tendinopathies vs. diffuse pain. Focal symptoms can be managed based on their standard treatment outside AIMSS, while care for diffuse symptoms focuses on the entire body. Among breast cancer survivors with AIMSS, aerobic exercise and supervised strength training have been found to decrease pain.⁷⁴ NSAIDs and acetaminophen have been used empirically. A randomized controlled trial found that duloxetine reduced pain in AIMSS.⁶⁷ Some supplements, such as glucosamine chondroitin, have shown improvements, and vitamin D with mixed results in pain and function was reported in certain studies.¹²⁸ Additionally, small studies have found that acupuncture and yoga can help improve AIMSS.^{31,54}

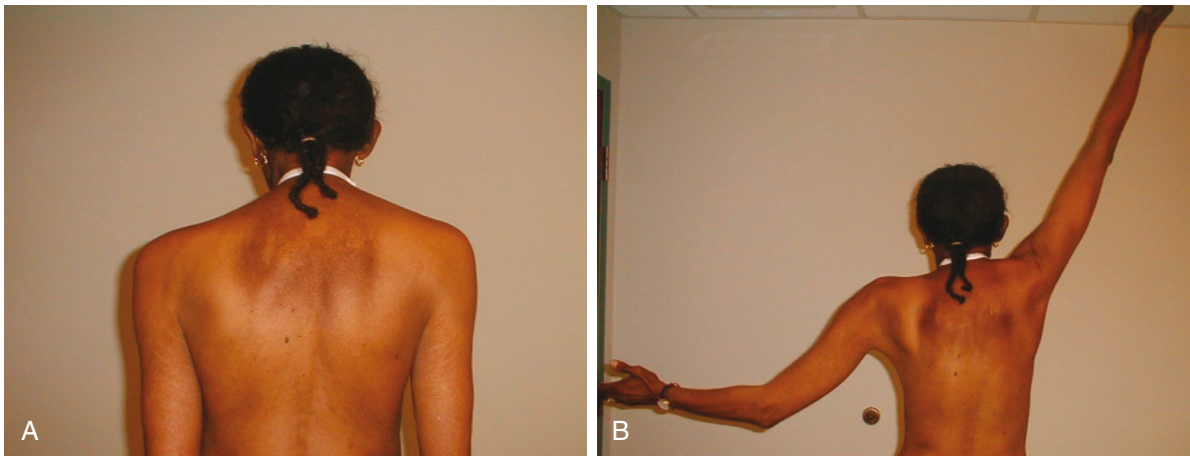
Head and Neck Cancer

Head and neck cancer is a diverse group of malignancies that can affect various areas, including the oral cavity, pharynx, larynx, salivary glands, and paranasal sinuses. It is the seventh most common cancer worldwide, and it is commonly diagnosed at an advanced stage, often requiring multimodal treatment.^{11,25} Survivors of head and neck cancer may experience a range of neuromusculoskeletal disorders, such as shoulder dysfunction, spinal accessory nerve palsy, cervical dystonia, dysphagia, dysarthria, trismus, lymphedema, and other conditions.¹²¹

Shoulder dysfunction is prevalent among head and neck cancer survivors, with reports of loss of ROM in abduction and pain as high as 100%.⁵⁵ In a small study, there were electrodiagnostic findings of spinal accessory nerve impairment in all cases of neck dissection, even when the spinal accessory nerve was intentionally spared.⁴⁵ Additionally, cervical nerve roots and the brachial plexus may be in the radiation field, resulting in weakness of the rotator cuff muscles.¹⁵⁴ This contributed to abnormal scapular and glenohumeral motion, negatively impacting shoulder function and pain.⁷² Fig. 30.6 shows an example of scapular dyskinesia. Rotator cuff tendinopathies and adhesive capsulitis are common.⁸³

The majority of head and neck cancer survivors experience neck dysfunction, with neck pain and decreased ROM. Cervical dystonia can be caused by injury to neuromuscular structures (spinal accessory nerve, cervical nerve roots, cervical plexus) and radiation fibrosis.¹⁵⁴ Patients may experience pain, tightness, and spasms in the neck muscles, particularly the sternocleidomastoid, scalene complex, and trapezius. Treatment options can include physical therapy with a lifelong home exercise program, medications for pain such as gabapentin or pregabalin, and botulinum toxin injections.^{28,154} Of note, appropriate muscle selection for botulinum toxin is key, and it is generally advised to avoid the posterior cervicothoracic musculature in this population due to the increased risk of overweakening, leading to dropped head syndrome.

Dropped head syndrome is characterized by weakness and atrophy of the cervicothoracic muscle leading to difficulty holding the head in a neutral position^{64,154} (Fig. 30.7). The prevalence of this for head and neck cancer survivors remains unknown. Although not needed for diagnosis, electrodiagnostic testing can reveal neuropathic, myopathic, or mixed results.⁵³ The cornerstone of treatment



• **Fig. 30.6** (A) Resting posture of a head and neck cancer patient with a complete spinal accessory nerve palsy. (B) Active shoulder abduction is limited to 90 degrees on the affected side, due to middle trapezius muscle weakness.



• **Fig. 30.7** Head-forward posture and exaggerated thoracic kyphosis, associated with radiation-induced soft tissue contracture, in a head and neck cancer patient.

is physical therapy that includes postural retraining, core strengthening, and stretching, especially of the pectoralis muscles, followed by a lifelong home exercise program. For more severe cases, a Head-master or Velpeau collar can be considered as an energy conservation device to improve posture ideally used as needed and not as a replacement of the home exercise program.¹⁵⁴

Trismus, a condition where the mouth opening is restricted to less than 35 mm, is experienced by nearly half of head and neck cancer survivors, and for one-third it can persist for years.^{40,177} This can deleteriously impact eating, oral hygiene, and quality of life and contribute to anxiety and depression.⁹⁶ Treatment options to improve jaw ROM include physical therapy and jaw stretching devices.^{29,139} Medication options include nerve stabilizing agents to address pain. Botulinum toxin injections into select muscles, including masseters and temporalis, can decrease pain and spasms.³³

Lymphedema within this population was historically underdiagnosed and treated, but newer studies reveal an incidence up to 90%. Swelling in this body region can also be internal, resulting in

swelling in the soft tissue and mucosa of aerodigestive tracts. Functional implications include potential detrimental effects on swallowing, voice, cervical ROM, body image, and overall quality of life. CDT remains the standard of care. Compression garments and advanced pneumatic compression devices can also be considered.^{131,147}

Brain Tumors

Brain tumors may be primary, originating from various cells within the CNS, or from other primary cancers that have metastasized to the CNS. Cancers that commonly spread to the CNS include lung, breast, colorectal, melanoma, and genitourinary cancers. Brain metastases are more prevalent than primary brain tumors, with an estimated 70,000 to 400,000 new cases annually in the United States.⁸⁷ Primary brain tumors make up less than 2% of all new cancer diagnoses, with the 2024 estimated number of new cases for brain and other CNS at 25,400; however, they cause significant disability and account disproportionately for morbidity and mortality.⁴

Individuals with brain tumors may experience wide ranging neurologic and functional deficits. Their presentation may vary depending on the size and location of the tumor. These functional impairments can be a direct result of the tumor itself, such as tissue destruction, compression of normal brain tissue, and increased intracranial pressure. They can also stem from the side effects of treatments, including tissue loss following surgery, steroid myopathy, and the acute or late effects from radiation therapy and chemotherapy. Potential symptoms include headache, mental disturbance, focal weakness, seizures, speech difficulty, visual disturbance, sensory disturbance, and gait or limb ataxia.^{73,168}

Advancements in surgical methods, chemotherapy, radiation therapy, and immunotherapy have extended survival time for individuals with brain tumors.⁷¹ Corticosteroids are commonly utilized in this population to reduce peritumor edema and brain compression. Given that brain tumors can cause considerable neurologic and functional impairments, it is crucial for patients to receive rehabilitation services to enhance their function and quality of life. A study revealed that over 80% of patients with CNS tumors display needs for rehabilitation, which were commonly multifactorial and included cognitive impairment, visual-perceptual

impairment, and weakness.^{91,114} Rehabilitation interventions are tailored to the individual's presentation, with goals akin to those for other brain disorders (e.g., optimizing functional transfers, bed mobility, ambulation, wheelchair training, family training). Equipment needs vary and may include an ambulatory assistive device, wheelchair, ankle-foot orthosis, reacher, long-handled shoehorn, and sock aids.¹²¹ Cognitive dysfunction and attentional deficits are common in individuals with brain tumors, significantly disrupting their quality of life. Cognitive therapy can help improve these areas and has been shown to enhance attention and verbal memory while also reducing mental fatigue, ultimately aiding in the social and professional reintegration of patients.¹²¹

Spinal Cord Tumors

Spinal cord tumors are categorized based on their anatomic location, either as extradural or intradural, with similar prevalence rates.¹²¹ Most extradural tumors are metastatic lesions, which are more common than primary spine tumors. Within the spine, the vertebral bodies are the most common site of metastasis. The cancers that frequently metastasize to the spine include those of the breast, lung, prostate, and kidney.⁴⁴ Epidural tumors are the most common cause of spinal cord injury (SCI) in cancer survivors. Leptomeningeal metastases occur in a small percentage of cancer survivors but can cause significant neurologic dysfunction. Leukemia, lymphoma, melanoma, breast, and lung cancers are the most common types that metastasize to the leptomeninges.³⁵

SCI can result from cancer itself, its treatment, or other preexisting or newly developed conditions. SCI can occur directly from the cancer, such as epidural, leptomeningeal, or intramedullary tumors, or indirectly from cancer treatment, such as radiation, chemotherapy, or surgery. Infections, vascular disorders, degenerative disorders, and metabolic disorders can also cause SCI in patients with cancer.

The initial symptoms of spinal tumors differ greatly based on their location, size, and the extent of spread. Intramedullary tumors often present with motor/sensory abnormalities and symptoms of myelopathy. Whether inside or outside the spinal cord, tumors can disrupt neural elements and pathways, resulting in both local and distant effects. Neurologic dysfunction is related to disruption of spinal cord pathways, leading to sensory changes, muscle weakness, difficulty in walking, and abnormalities in bowel, bladder, and sexual functioning. Metastatic epidural spinal cord compression is a medical emergency. Magnetic resonance imaging is typically the imaging modality of choice. This requires swift diagnosis and treatment, which may include corticosteroids, radiation therapy, and surgery.³⁰ Pain only during movement may indicate spinal instability, benefiting from surgical evaluation.⁶³

Individuals with a neoplastic SCI benefit from comprehensive rehabilitation services. Although life expectancy varies, the median survival for individuals with a neoplastic SCI was just over 4 months after the date of admission to rehabilitation.⁶¹ Expected functional recovery for neoplastic SCI tends to be less than that of nonneoplastic SCI. Acute SCI patient care may include managing pain, autonomic dysreflexia, spasticity, neurogenic bowel and bladder dysfunction, and ventilation for those with high cervical level injuries. There is also an increased risk of deep vein thrombosis, pulmonary embolism, infections, and pressure injuries.^{80,182}

Bone Metastasis

Bone is the most common site for metastatic spread. Cancers that commonly metastasize to the bone include lung, breast, prostate,

thyroid, renal cell, lymphoma, myeloma, and melanoma. Treatment of bone metastases may include medications, radiation therapy, and/or surgical stabilization depending on lesion location, pain, risk of fracture, and neurologic compromise as well as the patient overall prognosis and medical comorbidities. Most unfractured lesions are managed with systemic antineoplastic therapy or radiation therapy.⁹⁴

Bisphosphonates are utilized to decrease the progression of bone metastases and alleviate associated pain. Furthermore, bisphosphonates reduce the risk of fractures and hypercalcemia.¹³³ Radiation therapy offers an effective means of prompt local control of pain and tumor growth in bone metastases. Palliative radiation doses and protocols to bone lesions vary. Postsurgery radiation may be delayed but remains an adjunctive treatment to suppress tumor growth.⁴⁷

The greatest physiatrie concerns related to bony metastases are when those include lesions in the spine and long bones. These structures are crucial for weight bearing and mobility and are at risk of fracture. Painful osteolytic lesions are the lesions most prone to fracture. The clinical management of bone metastases at risk of fracture continues to be a challenge due to the difficulties in accurately determining the risk of fracture. Proposed rating scales include Mirels criteria for long bone lesions and spinal instability neoplastic score (SINS) for spine lesion. Details for scoring are outlined in [Tables 30.9](#) and [30.10](#). For Mirels criteria, a score greater than 9 suggests need for potential surgical fixation.^{99,110} A SINS score of 0 to 6 indicates a more stable lesions, while 7 to 12 is potentially unstable, and greater than 13 is unstable. Surgical consultation is recommended for those with a score of 7 or above.⁵⁰ In both risk assessments, functional pain is a significant predictor of potential pathologic fracture. However, an important note is that lesions are dynamic, and there may be tumor progression, which alters risk. For individuals with stable spines, treatment options may encompass oral and/or interventional pain management, physical therapies, and bracing. The goals of management of pathologic fractures include alleviating pain, maintaining or restoring function, stabilizing the bone, and controlling the local tumor.¹⁸⁰ Surgery is considered if the patient's life expectancy is more than 1 month for fractures in weight-bearing bones and more than 3 months for fractures in nonweight-bearing bones.¹⁸⁰

Precautions in Rehabilitation

Cancer survivors benefit from rehabilitation interventions; however, due to the medical complexity and impacts from tumor and/or treatments, they may require additional precautions to optimize safety during rehabilitation interventions.

Precautions should be taken for individuals with peripheral neuropathy, and these recommendations will be tailored to the individual patient and the severity of their condition. It is essential to remember that CIPN can fluctuate during treatment, often worsening with additional chemotherapy and improving with dose reduction or treatment completion or discontinuation. Neuropathy generally tends to be more sensory than motor, resulting in sensory ataxia rather than motor weakness, although both can occur. Balance training should be considered, along with aerobic alternatives, such as a stationary bike. Assistive devices and orthoses may be beneficial. Resistance exercises should also be considered, with a focus on safety. The hands should be monitored with the consideration of using gloves. Exercise machines may be safer than free weights. Given the increased risk of falls, gait and balance

TABLE 30.9 Spinal Instability Neoplastic Score

	Score
Location	
Junctional (occiput–C2, C7–T2, T11–L1, L5–S1)	3
Mobile spine (C3–C6, L2–L4)	2
Semirigid (T3–T10)	1
Rigid (S2–S5)	0
Pain	
Yes	3
Occasional pain but not mechanical	1
Pain-free lesion	0
Bone lesion	
Lytic	2
Mixed (lytic/blastic)	1
Blastic	0
Radiographic spinal alignment	
Subluxation/translation present	4
De novo deformity (kyphosis/scoliosis)	2
Normal alignment	0
Vertebral body collapse	
>50% collapse	3
<50% collapse	2
No collapse with >50% body involved	1
None of the above	0
Posterolateral involvement of spinal elements	
Bilateral	3
Unilateral	1
None of the above	0

TABLE 30.10 Mirels Criteria

Character	POINT ASSIGNED		
	1	2	3
Anatomic location	Upper extremity	Lower extremity	Trochanter
Lesion type	Blastic	Mixed	Lytic
Lesion size	≤1/3 of cortex	>1/3, <2/3	≥2/3
Pain	Mild	Moderate	Functional

should be continually assessed, and balance training should be incorporated into the patient's routine.¹⁵

Myelosuppression is a common side effect associated with chemotherapy, immunosuppressive agents, and corticosteroids and necessitates precautions. Rehabilitation services are crucial for this population, and studies suggest that exercise not only improves physical performance but also reduces the duration of anemia, neutropenia, thrombocytopenia, and the length of hospitalization.⁴¹ However, anemia can reduce exercise tolerance and endurance, resulting in fatigue, dizziness, and hemodynamic instability. A hemoglobin level of 8 g/dL has previously been used as a threshold for concern about exercise. However, the stability of the anemia and any recent drop in hemoglobin should also be taken into account. Rather than using fixed hemoglobin parameters to determine safe participation in rehabilitation therapies, it is recommended to adjust therapeutic activity based on patient symptoms such as dyspnea, lightheadedness, and chest pain.⁹⁹ Despite neutropenia, patients are encouraged to participate in therapy. There is no compelling evidence that rehabilitation interventions are contraindicated due to neutropenia. However, special consideration should be given to those experiencing side effects such as fatigue, malaise, dizziness, or lethargy. Infection precautions such as reverse isolation, avoiding public gyms and pools with a preference for home-based exercise, and practicing hand hygiene with antimicrobial products during every patient encounter are recommended.¹²³ The primary concern with thrombocytopenia is the increased risk of bleeding, which is especially important to consider for those undergoing active chemotherapy or receiving a bone marrow transplant. In addition to the absolute number of platelets, it is crucial to determine if the thrombocytopenia is relatively stable or if the platelet count is actively declining. There is some evidence that actively falling counts can be more concerning. Therefore it is important to collaborate closely with oncologists in managing this condition. General recommendations based on platelet counts are listed in [Box 30.5](#).¹¹²

When undergoing physical therapy, individuals with an ostomy need to take certain precautions to avoid complications such as a parastomal hernia. When performing resistance exercises, it is recommended to start with low intensity and gradually increase. The Valsalva maneuver, which involves holding one's breath and straining during a strenuous activity, should be avoided because it can increase intraabdominal pressure. Therapists should ensure that patients maintain appropriate form and technique during exercises to avoid creating unnecessary pressure on the abdomen. Core exercises may need to be modified to prevent excessive intraabdominal pressure. Individuals with an ileostomy are at an increased risk of dehydration, so it is crucial to stay hydrated before, during, and after exercise. Additionally, emptying the ostomy bag prior

• BOX 30.5 Precautions Based on Platelet Counts

- >80,000: Vigorous resistance and aerobic exercise
- 50,000 to 80,000: Progressive resistive exercise with activities such as stationary cycling
- 20,000 to 50,000: Minimal resistance (1 to 2 pounds) gentle active and passive range of motion, aerobic activity, ambulation, self-care
- 10,000 to 20,000: No resistance; gentle exercise, stretching, walking
- <10,000: limit activity

**TABLE
30.11****Overview of Modalities With Precautions and Contraindications**

Modality	Precautions	Contraindication
Heat	Impaired lymphatics, impaired sensation, scar tissue, wounds	Significantly impaired sensation, acutely irradiated tissue, peripheral vascular disease, active tumor
Cryotherapy	Impaired sensation, wounds, fragile skin	Ischemic tissue, peripheral vascular disease, Raynaud syndrome
Transcutaneous electrical nerve stimulation	Impaired sensation	Over active tumor, pacemaker, open wounds
Manual therapy	Impaired sensation, wounds, fragile skin	Over acutely irradiated tissue, active tumor, bone metastasis

to exercise can help prevent discomfort and potential accidents during physical activity.¹⁵

Bone metastasis and risk stratification assessments were previously discussed. Additional important considerations during physical therapy include precautions to avoid injury, especially in areas that are already compromised. High-impact activities, hyperflexion, hyperextension, and dynamic twisting should be avoided because they may put undue stress on affected sites. Exercise routines and guidance should be modified accordingly to account for these limitations. To alleviate pressure on the affected limb, weight-bearing activities can be offloaded using an assistive device. Fall prevention is crucial in therapy, and safety education is necessary, particularly regarding activities that require lifting to prevent fractures during activities of daily living. Functional pain should be regarded with caution because it can indicate a risk of fracture. Thus the patient should be educated about the importance of reporting discomfort immediately to the therapist.^{15,99}

Modalities including heat, cryotherapy, transcutaneous electrical nerve stimulation, and manual therapy can be utilized to help decrease pain. The cautions and contraindications are listed in Table 30.11, and these generally pertain to use directly over the tumor. Of note during end of life in the palliative care setting, their therapeutic modalities can be considered even in light of contraindications based on risk/benefit shared decision making based on the individuals' goals.^{99,153}

Summary

Cancer rehabilitation is an essential service that enables cancer survivors to achieve and sustain the highest possible levels of function.¹⁶⁰ Currently, cancer rehabilitation care and the providers available vary widely.²² For a truly holistic approach that addresses most functional issues, the rehabilitation team should ideally include physical therapy, occupational therapy, speech-language pathology, neuropsychology, cognitive rehabilitation, lymphedema therapy, pelvic floor therapy, and physical medicine and rehabilitation (physiatry). However, the reality is that this comprehensive team of specialists is seldom available to the survivors who require them. In fact, many patients face difficulties in accessing even a single proficient and informed practitioner, regardless of their specific needs.¹⁵⁷ The availability of effective treatment options and increase in survival rates highlight the importance of, value of, and need for more rehabilitation professionals to optimize and maintain cancer survivors' function and quality of life. Although there is substantial evidence supporting the role of cancer rehabilitation in enhancing function and quality of life across numerous conditions, there remains a substantial gap in high-level research to support this field's interventions.¹⁵³ Research studies demonstrating

benefit and value will be essential to promote integration and acceptance of comprehensive cancer rehabilitation as a standard in oncology care. Delivering comprehensive rehabilitation services benefits cancer survivors' function and quality of life.

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